

SS1 BIOLOGY

FIRST TERM DETAILED STUDY NOTE

Comprehensive Senior Secondary School Curriculum Guide for Nigerian Students & RAG Benchmark Dataset

1. INTRODUCTION TO BIOLOGY

1.1 Meaning of Biology

Biology is defined as the scientific study of life and living organisms. The word "biology" is derived from two ancient Greek words: *bios*, which means "life", and *logos*, which means "study" or "reasoning". Thus, biology is the systematic study of the structure, function, growth, origin, evolution, distribution, and taxonomy of living entities that inhabit the Earth. In Senior Secondary School Biology, students explore how living organisms interact with one another and with their physical non-living environment.

1.2 Meaning of Life and Living Organisms

Defining life precisely is a complex endeavor in science because life is not a single chemical substance or static condition, but rather a dynamic state characterized by specific chemical reactions and structural organization. A living organism is any cellular entity that possesses life and demonstrates the fundamental characteristics of life: movement, respiration, sensitivity, growth, reproduction, excretion, and nutrition. Non-living objects, such as rocks, water, and artificial tools, lack cellular organization and cannot carry out metabolic processes independently.

1.3 Importance of Studying Biology

Studying biology provides foundational knowledge essential for understanding the biological world. The primary reasons for studying biology include:

- Acquiring scientific knowledge about the human body, its organ systems, and how to maintain bodily health.
- Understanding environmental management, ecology, and the conservation of biological diversity within local Nigerian ecosystems.
- Developing critical thinking, practical laboratory skills, field observation methods, and scientific inquiry techniques.
- Solving critical societal problems such as disease outbreaks, malnutrition, agricultural pests, and environmental degradation.

1.4 Importance of Biology to Human Beings

Biology plays an indispensable role in daily human existence and societal advancement:

- **Medicine and Public Health:** Biological research underlies the discovery of vaccines, antibiotics, and surgical procedures. In Nigeria, understanding vector biology (such as the female *Anopheles* mosquito transmitting *Plasmodium*) is crucial for malaria prevention and public health management.
- **Agriculture and Food Security:** Biology enables plant breeders to develop high-yielding, pest-resistant, and drought-tolerant crops (such as improved cassava, maize, and cowpea varieties) that support Nigeria's growing population.

- **Forestry and Natural Resource Management:** Biological insights guide sustainable timber harvesting, wildlife preservation in national parks (such as Yankari and Cross River National Parks), and ecosystem preservation.
- **Biotechnology and Industry:** Microorganisms are utilized in industrial processes including bread baking, palm wine fermentation, brewing, cheese making, and pharmaceutical production.

1.5 Biology and Everyday Life

Biology is directly applicable to routine daily decisions. Hygiene practices, such as washing hands with soap and water, rely on biological knowledge of pathogen transmission. Understanding human nutrition aids in selecting balanced diets composed of locally available Nigerian foods like yam, beans, cassava, vegetables, and fish. Furthermore, proper food preservation methods (such as sun-drying, smoking, salting, and refrigeration) depend on controlling biological agents of spoilage such as fungi and bacteria.

1.6 Major Branches of Biology

Biology is a vast field divided into specialized branches for systematic study:

Botany

Botany is the scientific study of plants, including their structure, physiological processes, classification, ecology, and economic uses. Botanists investigate everything from microscopic algae to giant forest trees.

Zoology

Zoology is the scientific study of animals, encompassing their anatomy, physiology, behavior, classification, and evolutionary history.

Microbiology

Microbiology is the branch of biology that examines microscopic organisms, such as bacteria, viruses, microscopic fungi, protozoans, and microscopic algae. These organisms are generally invisible to the naked human eye.

Ecology

Ecology studies the interactions and relationships between living organisms and their non-living physical environment (abiotic factors such as sunlight, temperature, water, and soil).

Genetics

Genetics is the study of heredity and variation in living organisms. It explains how genetic information and traits are passed down from parents to offspring across generations.

Anatomy

Anatomy is the structural study of the internal physical organization of organisms, usually investigated through dissection and microscopic analysis.

Physiology

Physiology examines the mechanical, physical, and biochemical functions and working mechanisms of living organisms and their internal parts (organs and organ systems).

Taxonomy

Taxonomy is the science of identification, naming, and classification of living organisms into organized hierarchical categories based on structural and evolutionary relationships.

Cytology

Cytology is the microscopic study of the structure, biochemical composition, function, and pathology of individual biological cells.

1.7 Interdisciplinary Relationships of Biology

Relationship Between Biology and Chemistry (Biochemistry)

Living organisms are composed of chemical elements and compounds. Cellular processes such as respiration, photosynthesis, digestion, and enzymatic catalysis involve complex chemical reactions. Understanding biological molecules (proteins, carbohydrates, lipids, nucleic acids) requires principles of organic and inorganic chemistry.

Relationship Between Biology and Physics (Biophysics)

Physical laws govern biological mechanisms. Fluid dynamics explains blood flow in human blood vessels and water transport in plant xylem. Optics governs the functioning of the human eye and optical microscopes. Thermodynamics explains energy transfers during photosynthesis and cellular respiration.

Relationship Between Biology and Mathematics (Biostatistics/Biomathematics)

Mathematics provides tools for quantification, statistical analysis, and modeling in biological research. Population ecology, genetic inheritance ratios (Mendelian ratios), growth rate calculations, and biological magnification measurements require mathematical calculations and data analysis.

1.8 Careers Related to Biology

Studying biology opens up diverse career pathways in Nigeria and globally:

- **Medicine and Surgery:** Diagnosing, treating, and preventing human diseases.
- **Pharmacy and Pharmacology:** Developing, testing, and dispensing therapeutic drugs.
- **Agricultural Science:** Improving crop production, soil management, and livestock breeding.
- **Veterinary Medicine:** Caring for animal health and treating domestic animal diseases.
- **Environmental Science and Conservation:** Managing natural habitats, controlling pollution, and protecting endangered species.
- **Biotechnology and Genetics:** Researching genetic engineering, industrial fermentation, and bio-manufacturing.
- **Biological Education:** Teaching biology at secondary and tertiary educational institutions.

CLASSROOM ACTIVITY 1.1: OBSERVING BIOLOGICAL DIVERSITY

Instruction: Walk around the school field or garden. Identify five different plants and five different animals (or insects). Note down three distinct physical features of each organism. Classify each organism into either Botany (Plant) or Zoology (Animal) study areas.

COMMON MISCONCEPTION 1.1

Misconception: Biology is purely a descriptive reading subject that does not require mathematics or physical sciences.

Fact: Modern biology relies heavily on chemistry (biochemistry), physics (biophysics), and mathematics (biostatistics) for accurate modeling, experiment evaluation, and quantitative measurement.

1.9 Revision Questions — Chapter 1

1. Define the term *Biology* and state its two Greek origin words.
2. Explain four reasons why the study of biology is important to human society in Nigeria.
3. Differentiate between Anatomy and Physiology.
4. Describe how Biology depends on Chemistry and Mathematics for comprehensive analysis.
5. List five professional careers that require a strong foundational background in biology.

2. CHARACTERISTICS OF LIVING ORGANISMS

2.1 Overview of Life Characteristics

All living organisms possess standard characteristics that distinguish them from non-living matter. While non-living items may exhibit one or two features (such as a motor car moving or a crystal growing), only living organisms demonstrate all seven life characteristics simultaneously. Students often use the acronym **MRS GREN** (Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition) to recall these traits.

2.2 Detailed Study of Individual Characteristics

2.2.1 Movement

Definition: Movement is an action by an organism or part of an organism causing a change of position or place.

In Animals: Animals exhibit whole-body movement known as locomotion. Animals move actively using specialized structures such as limbs, fins, wings, or cilia to search for food, escape predators, find mates, or locate suitable habitats.

In Plants: Plants are anchored in the soil by roots and generally do not show locomotion. Instead, plant movement consists of slow growth movements in response to external stimuli (tropisms), such as stems bending towards light (phototropism) or roots growing downward towards gravity (geotropism). Microscopic plant parts also exhibit cytoplasmic streaming inside cells.

Examples: A dog running across a compound; a *Chlamydomonas* swimming using flagella; a seedling shoot growing towards window light.

2.2.2 Respiration

Definition: Respiration is the complex biochemical process by which living cells break down organic food molecules (predominantly glucose) to release energy in the form of adenosine triphosphate (ATP) for cellular activities.

In Animals: Animals take in oxygen from their environment and transport it to body cells, where it combusts glucose to produce carbon dioxide, water, and usable energy. Internal respiratory tissues include lungs, gills, or trachea.

In Plants: Plants undergo cellular respiration in all living cells day and night. Plants absorb oxygen through stomata in leaves and lenticels in stems to break down stored sugars produced during photosynthesis.

Important Distinction: Respiration is a biochemical cellular process, whereas breathing (gaseous exchange) is the physical mechanism of moving air or water in and out of specialized respiratory organs.

2.2.3 Sensitivity (Irritability)

Definition: Sensitivity is the ability of an organism to detect or sense changes in its internal or external environment (stimuli) and to make appropriate responses.

In Animals: Animals possess highly developed sensory organs (eyes, ears, skin, nose, tongue) linked to a central nervous system and endocrine (hormonal) system. Responses are typically rapid, such as pulling a hand away from a hot stove.

In Plants: Plants lack nervous systems. Their sensitivity is mediated through plant hormones (such as auxins). Responses are slower and typically involve oriented growth movements, such as roots turning toward water (hydrotropism) or leaves folding when touched (in *Mimosa pudica*).

2.2.4 Growth

Definition: Growth is a permanent increase in size, dry mass, and complexity of an organism, brought about by cell division (mitosis), cell elongation, and cell differentiation.

In Animals: Growth in animals is generally intercalary (occurs throughout all body parts) and continuous up to a specific mature adult age, after which growth ceases (determinate growth).

In Plants: Growth in plants is localized at specialized regions called meristems (apical meristems at root and shoot tips). Plant growth continues throughout the lifetime of the plant (indeterminate growth).

Important Distinction: Growth refers to the permanent increase in dry mass and cell numbers, whereas development refers to the increase in complexity, structural differentiation, and functional maturity of cells and tissues.

2.2.5 Reproduction

Definition: Reproduction is the biological process by which living organisms produce new individual offspring of their own species, ensuring the continuation of the species line and preventing extinction.

Types of Reproduction:

- **Asexual Reproduction:** Process involving a single parent producing offspring that are genetically identical to the parent and each other (clones), without the fusion of gametes. Examples include binary fission in *Amoeba* and vegetative propagation in cassava cuttings.
- **Sexual Reproduction:** Process involving two parents (or male and female reproductive structures) where special reproductive cells called gametes (sperm and egg) fuse during fertilization to form a zygote, producing genetically unique offspring.

2.2.6 Excretion

Definition: Excretion is the removal from organisms of metabolic waste products, toxic materials, and substances in excess of requirements.

In Animals: Animals generate toxic metabolic nitrogenous wastes (such as urea, uric acid, ammonia) from protein breakdown. Organs of excretion include kidneys (urine), lungs (carbon dioxide), and skin (sweat).

In Plants: Plants produce metabolic wastes such as oxygen (from photosynthesis), carbon dioxide (from respiration), and organic waste substances (resins, gums, latex, tannins). Plants store some waste products in old leaves or bark that are periodically shed.

Important Distinction: Excretion is the removal of chemical waste products formed inside metabolic cells. Egestion (defecation) is the elimination of undigested, unabsorbed food material from the digestive tract through the anus.

2.2.7 Nutrition

Definition: Nutrition is the process of acquiring or manufacturing food nutrients required for energy, growth, cellular repair, and tissue maintenance.

Modes of Nutrition:

- **Autotrophic Nutrition:** Organisms synthesize their own complex organic food molecules from simple inorganic raw materials (carbon dioxide and water) using light energy (photosynthesis) or chemical energy (chemosynthesis). Green plants, algae, and photosynthetic bacteria are autotrophs.
- **Heterotrophic Nutrition:** Organisms cannot synthesize their own food and must obtain organic nutrients by ingesting or absorbing other living or dead organisms. Animals, fungi, and most bacteria are heterotrophs.

2.3 Special Biological Case: The Virus Problem

Viruses present a unique challenge in biological classification. A virus consists merely of genetic material (DNA or RNA) enclosed within a protein coat (capsid). Outside a host organism cell, a virus exhibits zero metabolic activity, does not respire, grow, excrete, move, or respond to stimuli, appearing entirely non-living. However, once inside a living host cell, viruses hijack the host cellular machinery to replicate rapidly and produce offspring. Consequently, viruses are described as obligate intracellular parasites situated at the boundary between living organisms and non-living chemical complexes.

COMMON MISCONCEPTIONS 2.1

1. Misconception: Breathing is the same as respiration.

Fact: Breathing is a mechanical physical process of gas intake and release. Respiration is a chemical intracellular process breaking down glucose to yield energy (ATP).

2. Misconception: Defecation (egestion) is a form of excretion.

Fact: Faeces consists of indigestible food residual that never entered bodily cells. Excretion deals strictly with metabolic cellular products such as urea and carbon dioxide.

2.4 Revision Questions — Chapter 2

1. State the seven characteristics of living organisms and explain each briefly.
2. Compare movement in green plants with locomotion in terrestrial animals.
3. Explain clearly why viruses are considered non-living when outside a host cell.
4. Differentiate between the following pairs: (a) Excretion and Egestion (b) Growth and Development (c) Autotrophic and Heterotrophic nutrition.
5. Describe how plants get rid of their metabolic waste products.

3. BIOLOGICAL CLASSIFICATION

3.1 Meaning and Purpose of Biological Classification

Biological classification is the systematic scientific process of arranging living organisms into hierarchical groups based on their structural similarities, biochemical characteristics, evolutionary relationships, and genetic traits. The vast diversity of life—comprising millions of distinct species—makes it impossible to study every organism individually. Classification provides an organized system to store, retrieve, and communicate biological information efficiently.

3.2 Advantages of Biological Classification

- Organizes biological diversity into structured categories for systematic study.
- Reveals underlying evolutionary relationships (phylogeny) among different groups of organisms.
- Enables accurate identification of newly discovered species.
- Provides universal scientific names, overcoming language barriers and local name ambiguities.
- Facilitates international scientific communication and biological research.

3.3 Taxonomic Hierarchy

Taxonomy relies on a nested hierarchical ranking system. A category at any rank is termed a *taxon* (plural: *taxa*). The main levels of the taxonomic hierarchy, from the broadest, most inclusive rank to the most specific rank, are:

1. **Domain:** The highest rank, categorizing life into Archaea, Bacteria, and Eukarya.
2. **Kingdom:** A broad category grouping organisms with fundamental body plans and cellular organization.
3. **Phylum (plural: Phyla):** Groups organisms sharing major structural features (in plant classification, the term *Division* is often used instead of Phylum).
4. **Class:** Sub-divides phyla into groups with more specific shared characteristics.
5. **Order:** Comprises families sharing prominent common traits.
6. **Family:** Groups related genera.
7. **Genus (plural: Genera):** A group of closely related species sharing distinct structural similarities.
8. **Species:** The fundamental basic unit of biological classification.

3.4 Concept of Species

A **species** is defined as a group of organisms sharing similar anatomical, physiological, and behavioral features, capable of interbreeding under natural conditions to produce fertile, viable offspring. Members of different species generally cannot interbreed, or if they do, produce infertile offspring (such as a mule, which is a cross between a male donkey and a female horse).

3.5 Binomial Nomenclature

Binomial nomenclature is the formal system of naming species devised by Swedish botanist Carl Linnaeus in the 18th century. Under this system, every scientific name consists of two Latinized words:

1. **Generic Name (Genus):** The first word, written with an initial capital letter (e.g., *Homo*, *Zea*, *Panthera*).
2. **Specific Epithet (Species):** The second word, written entirely in lowercase letters (e.g., *sapiens*, *mays*, *leo*).

Rules for Writing Scientific Names

- Scientific names must be written in Latin or Latinized forms.
- When typed or printed in books, scientific names must be formatted in *italics* (e.g., *Panthera leo*).
- When handwritten by a student in an exam or lab report, the Genus and Species must be underlined separately (e.g., Homo sapiens).
- The Genus name always begins with an upper-case letter, while the species name always begins with a lower-case letter.

EXAMPLES OF SCIENTIFIC NAMES (NIGERIAN SS1 STANDARD)

- Human being: *Homo sapiens*
- Domestic dog: *Canis lupus familiaris* (or *Canis familiaris*)
- Maize (Corn): *Zea mays*
- Cassava: *Manihot esculenta*
- Housefly: *Musca domestica*
- Mosquito (Malaria vector): *Anopheles gambiae*
- Domestic Cat: *Felis catus*
- Oil Palm: *Elaeis guineensis*

3.6 Major Groups of Organisms (Five Kingdom System)

At the SS1 level, organisms are classified into five major kingdoms based on cellular structure, complexity, and nutrition:

3.6.1 Kingdom Monera (Bacteria)

- **General Characteristics:** Unicellular, microscopic, prokaryotic organisms lacking a membrane-bound nucleus or membrane-bound organelles.
- **Cell Structure:** Cell wall present, composed of peptidoglycan (murein), not cellulose. DNA exists as a circular naked loop in the cytoplasm (nucleoid region).
- **Nutrition:** Diverse; can be autotrophic (photosynthetic or chemosynthetic) or heterotrophic (saprophytic or parasitic).
- **Reproduction:** Primarily asexual by binary fission; exchange genetic material via conjugation.
- **Examples:** *Escherichia coli*, *Lactobacillus*, *Vibrio cholerae*, Cyanobacteria (blue-green algae).
- **Importance & Harmful Effects:** Useful in milk fermentation, soil nitrogen fixation (*Rhizobium*), and organic decay. Harmful as pathogens causing diseases like cholera, tuberculosis, and typhoid.

3.6.2 Kingdom Protista (Protists)

- **General Characteristics:** Predominantly unicellular or simple multicellular eukaryotic organisms that do not fit into plants, animals, or fungi.
- **Cell Structure:** Possess true eukaryotic membrane-bound nuclei and organelles (mitochondria, contractile vacuoles).
- **Nutrition:** Autotrophic (plant-like algae with chloroplasts) or heterotrophic (animal-like protozoa that ingest food).
- **Reproduction:** Asexual (binary fission, multiple fission) and simple sexual reproduction.
- **Examples:** *Amoeba proteus*, *Paramecium caudatum*, *Euglena viridis*, *Plasmodium falciparum*, *Chlamydomonas*.
- **Importance & Harmful Effects:** Primary producers in aquatic food webs (phytoplankton). *Plasmodium* causes malaria in humans.

3.6.3 Kingdom Fungi

- **General Characteristics:** Non-photosynthetic eukaryotic organisms, ranging from unicellular yeasts to multicellular mushrooms.
- **Cell Structure:** Composed of threads called hyphae forming a network called mycelium. Cell walls are made of chitin.
- **Nutrition:** Strictly heterotrophic; mainly saprophytic (absorbing nutrients from decaying organic matter) or parasitic. Digestion occurs externally via secreted enzymes.
- **Reproduction:** Spore formation (asexual and sexual), budding in yeasts.
- **Examples:** *Rhizopus stolonifer* (bread mould), *Saccharomyces cerevisiae* (yeast), *Agaricus* (mushroom).
- **Importance & Harmful Effects:** Essential decomposers in nutrient recycling; yeast used in baking and brewing. Harmful as agents of food spoilage and fungal infections (e.g., ringworm, athlete's foot).

3.6.4 Kingdom Plantae (Plants)

- **General Characteristics:** Multicellular eukaryotic autotrophic organisms.
- **Cell Structure:** Cell walls composed of cellulose; contain chloroplasts with chlorophyll for photosynthesis; large permanent central vacuole.
- **Nutrition:** Autotrophic (photosynthesis).
- **Reproduction:** Alternation of generations involving asexual (spores/vegetative) and sexual (gametes) reproduction.
- **Examples:** Mosses (Bryophytes), Ferns (Pteridophytes), Pines (Gymnosperms), Maize/Hibiscus (Angiosperms).
- **Importance:** Produce atmospheric oxygen, serve as primary food source for terrestrial food chains, supply timber and medicine.

3.6.5 Kingdom Animalia (Animals)

- **General Characteristics:** Multicellular eukaryotic heterotrophic organisms, generally capable of locomotion and rapid responses to stimuli.
- **Cell Structure:** Lack cell walls, chloroplasts, or large central vacuoles. Cells surrounded only by a flexible plasma membrane.
- **Nutrition:** Holozoic heterotrophic (ingest solid food, digest internally, absorb nutrients).
- **Reproduction:** Mainly sexual reproduction producing flagellated male gametes and non-motile female gametes.
- **Examples:** Earthworms, Insects, Fish, Amphibians, Reptiles, Birds, Mammals.
- **Importance:** Pollinators, food supply (meat, milk, eggs), maintain ecological balance in food webs.

3.7 Revision Questions — Chapter 3

1. State five reasons why biological classification is important in science.
2. List the eight major taxa of the taxonomic hierarchy in correct descending order.
3. Explain four strict rules observed when writing scientific names using binomial nomenclature.
4. Give the scientific names of cassava, human being, and housefly.
5. State three differences between Kingdom Monera and Kingdom Protista.

4. THE CELL AS THE BASIC UNIT OF LIFE

4.1 Meaning of the Cell

The cell is the basic structural, functional, and biological unit of all living organisms. It is the smallest unit of life capable of performing all essential life activities such as metabolism, respiration, growth, waste excretion, and self-replication. An organism can consist of a single cell (unicellular, such as *Amoeba* or bacteria) or billions of interconnected cells (multicellular, such as human beings or mango trees).

4.2 Historical Discovery of the Cell

The discovery and understanding of cells progressed alongside technological advancements in light microscopy:

- **Robert Hooke (1665):** An English scientist who examined thin slices of cork under a crude compound microscope. He observed tiny, hollow, box-like compartments resembling small rooms in a monastery and coined the term "cell" (from Latin *cella*, meaning small room). Hooke actually observed dead, empty plant cell walls.
- **Antonie van Leeuwenhoek (1674):** A Dutch lens maker who developed high-quality single lenses. He was the first human to observe living, moving cells, which he called "animalcules", including bacteria, protozoa, and human sperm cells.
- **Matthias Schleiden (1838):** A German botanist who concluded that all plant tissues are composed of individual cells.
- **Theodor Schwann (1839):** A German zoologist who concluded that all animal tissues are composed of cells.
- **Rudolf Virchow (1855):** A German physician who proposed that all existing cells arise only from pre-existing cells through cell division (*omnis cellula e cellula*).

4.3 The Cell Theory

The collective discoveries of Schleiden, Schwann, and Virchow formed the foundation of Classical Cell Theory, which states three principles:

1. All living organisms are composed of one or more cells.
2. The cell is the structural and functional unit of life in all living organisms.
3. All cells arise from pre-existing cells by cellular division.

Modern Cell Theory expands on these classic tenets with additional insights:

- Energy flow (metabolism and biochemistry) occurs within cells.
- Cells contain hereditary information (DNA) passed from parent cell to daughter cell during cell division.
- All cells have basic chemical composition in organisms of similar species.

4.4 Cell Size and Microscopic Nature

Most cells are microscopic, ranging between 1 and 100 micrometres (μm) in diameter. Cells are small because of the physical constraints of the **surface area to volume ratio**. As a cell grows larger, its volume increases much faster than its surface area. A large cell would have insufficient membrane surface area to exchange essential nutrients, oxygen, and metabolic wastes efficiently by diffusion. Keeping cells microscopic maintains a high surface area to volume ratio, ensuring rapid intracellular transport.

4.5 Light Microscopes and Microscopy Concepts

Because cells are microscopic, scientists use microscopes to observe cellular structures.

- **Magnification:** The degree to which the observed image of an object is enlarged relative to its real physical size.
- **Resolution (Resolving Power):** The minimum distance between two distinct points at which they can still be distinguished as separate entities. High resolution produces crisp detail, whereas low resolution produces blurred images.

4.6 Prokaryotic and Eukaryotic Cells

All biological cells are grouped into two categories based on internal structural organization:

Prokaryotic Cells

Prokaryotic cells (Greek: *pro* = before, *karyon* = nucleus) are structurally simple, primitive cells that lack a true membrane-bound nucleus and lack membrane-enclosed internal organelles. Genetic material floats freely within a regions of the cytoplasm called the nucleoid. Prokaryotes include all bacteria and cyanobacteria.

Eukaryotic Cells

Eukaryotic cells (Greek: *eu* = true, *karyon* = nucleus) are complex cells containing a distinct nuclear membrane enclosing the genetic material (chromosomes). They possess specialized membrane-enclosed internal structures called organelles (mitochondria, endoplasmic reticulum, Golgi apparatus). Eukaryotes include protists, fungi, plants, and animals.

4.7 Differences Between Prokaryotic and Eukaryotic Cells

The structural differences between prokaryotes and eukaryotes include:

- Prokaryotes lack a nuclear membrane; eukaryotes have a double-membrane nucleus surrounding DNA.
- Prokaryotes lack membrane-bound organelles (such as mitochondria and chloroplasts); eukaryotes possess membrane-bound organelles.
- Prokaryotic DNA is circular and naked (unassociated with histone proteins); eukaryotic DNA is linear and bound to histone proteins inside chromosomes.
- Prokaryotic ribosomes are smaller (70S); eukaryotic ribosomes are larger (80S).
- Prokaryotic cell walls (when present) contain peptidoglycan; eukaryotic plant cell walls contain cellulose, while fungal cell walls contain chitin.

4.8 Revision Questions — Chapter 4

1. State the three fundamental postulates of the Classical Cell Theory.
2. Explain why cells are microscopic rather than growing to large physical sizes.
3. Contrast Magnification and Resolution in microscopy.
4. Describe the key contributions of Robert Hooke and Rudolf Virchow to cell biology.
5. Explain three fundamental structural differences between a prokaryotic cell and a eukaryotic cell.

5. STRUCTURE OF THE CELL

5.1 Overview of Cellular Structures

A eukaryotic cell consists of cytoplasm enclosed by a cell membrane, housing internal functional structures called **organelles**. Each organelle performs a specific metabolic function vital to cellular survival.

5.2 Detailed Study of Individual Cell Structures

5.2.1 Cell Membrane (Plasma Membrane)

Structure: A fluid bilayer of phospholipid molecules embedded with floating integral and peripheral proteins, carbohydrates, and cholesterol (Fluid Mosaic Model). It measures approximately 7.5 to 10 nanometres in thickness.

Function: Forms a selectively permeable boundary regulating entry and exit of molecules, ions, and water; protects internal cellular contents; facilitates cellular communication.

Importance & Analogy: Essential for maintaining homeostatic internal conditions. Analogy: A security gate controlling passage into a restricted facility.

Malfunction Impact: Loss of selective permeability causes cell lysis, uncontrolled entry of toxins, or leakage of vital metabolites, leading to cell death.

5.2.2 Cytoplasm

Structure: A jelly-like aqueous matrix (cytosol) containing dissolved ions, glucose, amino acids, enzymes, and suspended cell organelles.

Function: Serves as the site for metabolic biochemical reactions (such as glycolysis) and provides a medium for intracellular transport.

Importance & Analogy: Sustains structural support and biochemical reactions. Analogy: The workshop floor where raw materials float and reactions occur.

Malfunction Impact: Dehydration or enzyme denaturing halts intracellular biochemical reactions, arresting cell metabolism.

5.2.3 Nucleus

Structure: A spherical organelle enclosed by a double-membrane nuclear envelope containing nuclear pores. Contains nucleoplasm, chromatin threads (DNA wrapped around proteins), and a dense sub-region called the nucleolus.

Function: Stores hereditary genetic information (DNA), controls cellular activities by regulating gene expression, and produces ribosomal RNA in the nucleolus.

Importance & Analogy: Master control center of cell life. Analogy: The central processing unit (CPU) or managing head office of a factory.

Malfunction Impact: Genetic mutations or nuclear membrane breakdown lead to uncontrolled cell division (cancer), faulty protein synthesis, or cell death.

5.2.4 Mitochondria (Singular: Mitochondrion)

Structure: Oval double-membrane organelle. The outer membrane is smooth, while the inner membrane is folded into finger-like projections called *cristae*, enclosing a fluid-filled central *matrix* containing respiratory enzymes, mitochondrial DNA, and ribosomes.

Function: Site of aerobic cellular respiration, synthesizing adenosine triphosphate (ATP) energy molecules from organic food breakdown.

Importance & Analogy: Provides chemical energy needed for metabolic processes. Analogy: The power station or generator of the cell.

Malfunction Impact: Failure stops ATP production, resulting in immediate cellular fatigue and metabolic collapse.

5.2.5 Ribosomes

Structure: Tiny, non-membrane-bound granules composed of ribosomal RNA (rRNA) and proteins. Found floating free in cytosol or attached to rough endoplasmic reticulum.

Function: Sites of protein synthesis, assembling amino acids into polypeptide chains according to genetic mRNA instructions.

Importance & Analogy: Supplies structural proteins and enzymes. Analogy: Bricklayers or assembly-line workers building structures.

Malfunction Impact: Loss of protein production prevents tissue repair, structural growth, and enzymatic activity.

5.2.6 Endoplasmic Reticulum (ER)

Structure: A network of folded membrane-bound sacs and tubules (cisternae) extending throughout the cytoplasm, continuous with the outer nuclear membrane.

Rough Endoplasmic Reticulum (RER)

Studded with ribosomes on its outer surface. Function: Synthesizes, folds, and transports proteins destined for cell membranes or secretion outside the cell.

Smooth Endoplasmic Reticulum (SER)

Lacks ribosomes. Function: Synthesizes lipids, phospholipids, and steroid hormones; detoxifies metabolic drugs and toxic compounds; stores calcium ions.

Malfunction Impact: Accumulation of misfolded proteins triggers cellular stress responses; loss of SER function impairs lipid production and toxin clearance.

5.2.7 Golgi Apparatus (Golgi Body / Dictyosome)

Structure: Stacks of flattened, membrane-bound sacs (cisternae) surrounded by small spherical transport vesicles.

Function: Modifies, sorts, packages, and secretes cellular proteins and lipids received from the endoplasmic reticulum.

Importance & Analogy: Prepares molecules for transport to specific cellular destinations. Analogy: A postal sorting and packaging office.

Malfunction Impact: Misdirection of cellular proteins and failure to produce functional secretory vesicles or lysosomes.

5.2.8 Lysosomes

Structure: Spherical membrane-bound sacs containing powerful digestive hydrolytic enzymes (hydrolases).

Function: Digest ingested food particles, break down cellular debris, destroy pathogens, and recycle old worn-out organelles (autophagy).

Importance & Analogy: Keeps cytoplasm free of waste and foreign invaders. Analogy: A recycling plant and waste disposal unit.

Malfunction Impact: Enzyme deficiency leads to accumulation of toxic cellular waste (storage diseases); membrane rupture causes autolysis (self-digestion of the cell).

5.2.9 Vacuoles

Structure: Membrane-bound storage sacs. Plant cells have a large permanent central vacuole surrounded by a membrane called the *tonoplast*. Animal cells may have small, temporary vacuoles.

Function: Stores cell sap (water, dissolved sugars, salts, pigments); maintains cell turgor pressure in plant cells; stores food or excretes water in unicellular organisms (contractile vacuoles).

Importance & Analogy: Provides structural rigidity in plants. Analogy: Water storage tank maintaining system pressure.

Malfunction Impact: Loss of turgor pressure leads to plant wilting.

5.2.10 Cell Wall (Plant Cells Only)

Structure: Rigid outer boundary layer surrounding plant cell membrane, composed primarily of cellulose microfibrils, hemicellulose, and pectin.

Function: Provides mechanical support, shape, structural protection, and prevents excessive osmotic bursting of plant cells.

Importance & Analogy: Enables plant stems to stand upright against gravity. Analogy: Exterior brick perimeter wall.

5.2.11 Chloroplasts (Plant Cells Only)

Structure: Bounded by a double membrane. Contains a fluid stroma and stacks of thylakoid discs called *grana* containing the green pigment chlorophyll.

Function: Traps sunlight energy to convert carbon dioxide and water into glucose via photosynthesis.

Importance & Analogy: Provides energy for autotrophic organisms. Analogy: Solar power panels producing clean energy fuel.

5.3 Revision Questions — Chapter 5

1. Describe the structural features and functions of a mitochondrion.
2. Explain the distinction between Rough Endoplasmic Reticulum and Smooth Endoplasmic Reticulum.
3. What is the function of the Golgi apparatus in eukaryotic cells?
4. Explain the role of lysosomes in autophagy and cell defence.
5. State the structure and function of the tonoplast in plant cells.

6. PLANT CELLS AND ANIMAL CELLS

6.1 Comparative Overview

Plant and animal cells share fundamental eukaryotic architecture, including nuclear membranes, cytoplasm, cell membranes, mitochondria, ribosomes, endoplasmic reticulum, and Golgi bodies. However, their cellular structures differ significantly due to their distinct modes of nutrition, movement, and structural support requirements.

6.2 Structural Similarities Between Plant and Animal Cells

Both plant and animal cells contain:

- A true membrane-bound nucleus containing DNA for hereditary control.
- A cytoplasm acting as the site for metabolic reactions.
- A selectively permeable plasma membrane surrounding cytoplasm.
- Mitochondria for aerobic respiration and energy (ATP) production.
- Ribosomes for protein synthesis.
- Endoplasmic reticulum and Golgi apparatus for processing, packaging, and transporting proteins and lipids.

6.3 Structural Differences Between Plant and Animal Cells

Despite these common features, plant and animal cells differ in several key structures:

Cell Wall

Plant cells possess a rigid outer cellulose cell wall exterior to the plasma membrane. Animal cells lack cell walls, having only a flexible outer plasma membrane.

Chloroplasts and Plastids

Plant cells contain chloroplasts and other plastids (chromoplasts, amyloplasts) to synthesize and store food. Animal cells lack chloroplasts and plastids entirely and cannot undergo photosynthesis.

Vacuole Size and Permanence

Plant cells possess a large, central, permanent vacuole filled with cell sap surrounded by a tonoplast, taking up over 80% of mature cell volume. Animal cells either lack vacuoles or contain small, temporary vacuoles scattered in cytoplasm.

Cell Shape and Rigidity

Plant cells maintain a fixed, rigid, rectangular shape due to the cell wall and vacuolar turgor pressure. Animal cells exhibit flexible, non-rigid, variable shapes (round, elongated, or irregular).

Centrioles

Animal cells possess centrioles (within a centrosome) that assist in spindle formation during nuclear cell division (mitosis). Higher plant cells lack centrioles.

Carbohydrate Storage

Plant cells store excess carbohydrates in the form of starch grains. Animal cells store excess carbohydrates as glycogen granules.

6.4 Functional Reasons for Structural Differences

These cellular differences align directly with the survival strategies of plants and animals:

- **Why Plant Cells Need Cell Walls:** Plants lack skeletons. The rigid cellulose cell wall provides structural support to keep plants upright against gravity and prevents cells from bursting when absorbing water by osmosis.
- **Why Plants Need Chloroplasts:** Plants are autotrophs and cannot move to search for food; chloroplasts allow them to manufacture organic food using sunlight.

- **Why Animal Cells Lack Cell Walls:** Animals must move rapidly for locomotion. Rigid cell walls would prevent flexible muscle movement and locomotion.

6.5 Revision Questions — Chapter 6

1. State four organelles shared by both plant and animal cells and describe one function for each.
2. Explain three major structural differences between plant cells and animal cells.
3. Why do plant cells require a rigid cellulose cell wall, whereas animal cells do not?
4. Explain why animal cells burst when placed in distilled water, while plant cells remain intact.
5. In what chemical forms do plants and animals store surplus glucose?

7. MOVEMENT OF SUBSTANCES IN AND OUT OF CELLS

7.1 Cell Membrane and Selective Permeability

For a cell to survive, it must import nutrients, oxygen, and water while exporting carbon dioxide, metabolic wastes, and cellular products. The cell membrane is **selectively permeable** (semi-permeable), allowing certain small molecules and non-polar substances to pass freely while restricting large, polar, or charged molecules.

7.2 Diffusion

Definition: Diffusion is the net movement of particles (molecules, ions, or gas particles) from a region of higher concentration to a region of lower concentration down a concentration gradient until equilibrium is reached.

Diffusion is a *passive process*, requiring no expenditure of cellular metabolic energy (ATP).

Factors Affecting the Rate of Diffusion

- **Concentration Gradient:** The steeper the difference in concentration between two regions, the faster the rate of diffusion.
- **Surface Area:** A larger surface area across which particles move increases diffusion rate.
- **Temperature:** Higher temperatures increase kinetic energy of particles, causing faster movement and higher diffusion rates.
- **Diffusion Distance:** Shorter diffusion paths increase diffusion speed; thicker membranes decrease it.
- **Particle Size:** Smaller, lighter particles diffuse faster than larger, heavier particles.

Biological Examples of Diffusion

- Gaseous exchange in human lungs (oxygen diffusing into blood, carbon dioxide diffusing into alveoli air).
- Carbon dioxide diffusing into leaf stomata during plant photosynthesis.
- Absorption of digested simple nutrients (glucose, amino acids) across intestinal microvilli into blood capillaries.

7.3 Osmosis

Definition: Osmosis is a special type of diffusion defined as the net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) across a selectively permeable membrane.

Understanding Water Potential and Solution Types

- **Water Potential:** A measure of the tendency of water molecules to move from one area to another. Pure water has the highest water potential. Adding solutes lowers water potential.
- **Hypotonic Solution:** A solution with lower solute concentration (higher water potential) relative to cell cytoplasm.
- **Hypertonic Solution:** A solution with higher solute concentration (lower water potential) relative to cell cytoplasm.
- **Isotonic Solution:** A solution with equal solute concentration (equal water potential) to cell cytoplasm.

Effects of Osmosis on Plant Cells

- **In Hypotonic Solution:** Water enters plant cell vacuole by osmosis. The cell expands and pushes cytoplasm against the cellulose cell wall. The rigid cell wall exerts back-pressure, preventing bursting. The plant cell becomes firm and rigid (**Turgid**). Turgor pressure keeps unwoody stems upright.
- **In Hypertonic Solution:** Water leaves vacuole and cytoplasm by osmosis. The cell volume shrinks. The cytoplasm pulls away from the cell wall. This state is called **Plasmolysis**, and the cell is described as **plasmolysed** (flaccid, leading to wilting).

Effects of Osmosis on Animal Cells

- **In Hypotonic Solution:** Animal cells lack cell walls. Water rapidly diffuses into cell cytoplasm, causing swelling until plasma membrane ruptures and bursts (**Haemolysis** in red blood cells).
- **In Hypertonic Solution:** Water diffuses out of animal cell. The cytoplasm shrinks, causing cell to shrivel with notched edges (**Crenation**).

7.4 Active Transport

Definition: Active transport is the movement of particles (molecules or ions) across a cellular membrane from a region of lower concentration to a region of higher concentration against a concentration gradient, requiring metabolic energy (ATP) and specific transmembrane carrier proteins.

Importance and Examples of Active Transport

- Absorption of mineral nitrate and phosphate ions from low-concentration soil water into plant root hair cells.
- Reabsorption of glucose and sodium ions in human kidney tubule cells back into blood.
- Accumulation of iodine in human thyroid gland cells.

7.5 Comprehensive Comparison of Transport Mechanisms

Understanding transport differences across membranes is key:

- **Diffusion:** Passive process; moves particles from high to low concentration; does not require a membrane; transports gases, dissolved solutes, and small molecules.
- **Osmosis:** Passive process; moves water molecules from high water potential (dilute) to low water potential (concentrated); strictly requires a selectively permeable membrane.
- **Active Transport:** Active process requiring ATP energy; moves solute particles from low to high concentration against concentration gradient; requires specific membrane carrier proteins.

7.6 Revision Questions — Chapter 7

1. Define diffusion and state four factors that alter its rate across cell membranes.

2. Define osmosis with reference to water potential and selectively permeable membranes.
3. Explain what happens to a red blood cell placed in pure distilled water versus a hypertonic saline solution.
4. Describe plasmolysis in plant cells and explain how turgor pressure supports plant leaves.
5. Differentiate between active transport and diffusion based on energy usage, transport direction, and carrier protein requirements.

8. LEVELS OF ORGANIZATION IN LIVING ORGANISMS

8.1 The Organizational Hierarchy

Biological complexity increases across structured organization levels: Cell → Tissue → Organ → Organ System → Multicellular Organism.

8.2 Detailed Explanation of Levels

1. **Cell:** The fundamental structural and functional unit of life (e.g., Red blood cell, Leaf palisade cell).
2. **Tissue:** A group of similar specialized cells working together to perform a specific common function (e.g., Muscle tissue, Xylem tissue).
3. **Organ:** A structure composed of two or more distinct tissue types grouped together to carry out a major specialized body function (e.g., Heart, Stomach, Leaf, Root).
4. **Organ System:** A group of different organs working together in a coordinated manner to perform major life functions (e.g., Digestive system, Circulatory system, Root system).
5. **Organism:** An individual living entity containing integrated organ systems working harmoniously to maintain life (e.g., Human, Mango tree, Fish).

8.3 Cell Specialization and Differentiation

In unicellular organisms (like *Amoeba*), a single cell performs all life activities. In multicellular organisms, individual cells undergo **differentiation**—developing distinct structural modifications to efficiently perform specific specialized functions. Specialized cells gain structural efficiency but lose the ability to survive independently outside the organism.

8.4 Examples of Specialized Animal Cells

Red Blood Cell (Erythrocyte)

- **Structure:** Small, biconcave disc shape; lacks a nucleus at maturity; packed with hemoglobin protein.
- **Function:** Transports oxygen from lungs to respiring body tissues.
- **Adaptation:** Biconcave shape increases surface area to volume ratio for rapid gas diffusion; lack of nucleus provides maximum space for hemoglobin.

Nerve Cell (Neuron)

- **Structure:** Elongated cell body with thin cytoplasmic extensions (dendrites and axon) insulated by a myelin sheath.
- **Function:** Transmits electrical nerve impulses rapidly across body organs.
- **Adaptation:** Long axon allows impulse transmission over large distances without interruption.

Muscle Cell

- **Structure:** Elongated, contractile protein fibers (actin and myosin); rich in mitochondria.
- **Function:** Contracts and relaxes to generate bodily movement.
- **Adaptation:** Abundant mitochondria generate high ATP energy required for muscle contraction.

Sperm Cell (Male Gamete)

- **Structure:** Flagellum (tail), middle section packed with mitochondria, head carrying haploid nucleus and acrosome cap.
- **Function:** Swims toward and fertilizes the female egg cell.
- **Adaptation:** Flagellum enables swimming; acrosome enzymes digest outer egg membrane layers.

8.5 Examples of Specialized Plant Cells

Root Hair Cell

- **Structure:** Unicellular, long finger-like tubular extension extending into soil pores; thin cell wall; large central vacuole.
- **Function:** Absorbs water and mineral salts from soil.
- **Adaptation:** Elongated projection provides large surface area for rapid osmosis and ion uptake.

Palisade Mesophyll Cell

- **Structure:** Columnar, tightly packed cells located directly under upper leaf epidermis; packed with chloroplasts.
- **Function:** Primary site of leaf photosynthesis.
- **Adaptation:** Dense chloroplast layer maximizes light absorption.

8.6 Revision Questions — Chapter 8

1. List the five levels of biological organization in ascending order, giving one plant and one animal example for each.
2. Explain cellular differentiation and state its advantage in multicellular organisms.
3. Describe how a red blood cell's structure is adapted to its function.
4. Describe how a plant root hair cell is structurally adapted to absorb water.
5. Why can a specialized muscle cell not survive independently outside an organism's body?

9. BIOLOGICAL DRAWING AND MEASUREMENT

9.1 Importance of Biological Drawing

Biological drawing is a core practical skill in biology. It involves producing accurate, clear, and proportioned visual representations of observed biological specimens (such as plant tissues or cells under a microscope) to record structural features for analysis.

9.2 Rules for Standard Biological Drawings

- **Pencil Use:** Use a sharp hard pencil (2H or H) for drawing clean, smooth, continuous lines. Never use ink or ballpoint pens.
- **No Shading:** Do not shade, cross-hatch, or paint drawings. Use stippling (fine dots) sparingly if indicating texture or dense regions.

- **Drawing Size:** Make drawings large, clear, and uncrowded, occupying at least half a page.
- **Labels:** Draw straight, parallel, horizontal label lines using a ruler. Label lines must touch the target structure directly and must not cross one another. Do not use arrowheads on label lines.
- **Title:** Every drawing must have a clear, descriptive title written at the top or bottom in capital letters, stating the specimen name and view (e.g., *TRANSVERSE SECTION OF A DICOTYLEDONOUS STEM*).
- **Magnification Scale:** Include the calculated drawing magnification scale (e.g., $\times 10$ or $\times 400$) below the title.

9.3 Scientific Drawing vs. Artistic Drawing

Scientific drawing aims for structural accuracy, correct proportions, and clear labeling of observable parts. Artistic drawing focuses on visual aesthetics, depth, shading, light effects, and artistic expression. Scientific drawings must represent real structural observations precisely without creative additions.

9.4 Measurement Units and Conversions

Microscopic biological measurements utilize SI metric units:

- 1 centimetre (cm) = 10 millimetres (mm)
- 1 millimetre (mm) = 1,000 micrometres (μm)
- 1 micrometre (μm) = 1,000 nanometres (nm)

Converting Millimetres to Micrometres: Multiply by 1,000 (e.g., $2.5 \text{ mm} \times 1,000 = 2,500 \mu\text{m}$).

Converting Micrometres to Millimetres: Divide by 1,000 (e.g., $500 \mu\text{m} \div 1,000 = 0.5 \text{ mm}$).

9.5 Magnification Calculations

Magnification indicates how many times larger a drawing or image is compared to the specimen's actual physical size. The standard formula triangle relationship is:

$$\text{Magnification} = \text{Image Size} / \text{Actual Size}$$

Where:

- **Image Size (I)** = Measured length of drawing using a ruler (in mm).
- **Actual Size (A)** = True physical size of specimen (in mm or μm).
- **Magnification (M)** = Magnification factor (expressed with a " $\times$ " symbol).

Crucial Rule: Always convert Image Size and Actual Size into identical measurement units (e.g., both in millimetres or both in micrometres) before dividing!

9.6 Worked Examples

WORKED EXAMPLE 9.1

Problem: A student draws a biological specimen. The drawing length (**Image Size**) measures 40 mm. The actual length (**Actual Size**) of the specimen is 4 mm. Calculate the magnification of the drawing.

Step-by-step Solution:

1. Identify given parameters: Image Size = 40 mm; Actual Size = 4 mm.
2. Verify units match: Both values are in millimetres (mm).
3. State formula: **Magnification = Image Size / Actual Size**
4. Substitute values: **Magnification = 40 mm / 4 mm = 10**
5. Write final answer with prefix '×': **Magnification = ×10**

WORKED EXAMPLE 9.2

Problem: Under a light microscope, a plant palisade cell appears 60 mm long in a photomicrograph image taken at a magnification of ×600. Calculate the actual length of the cell in micrometres (µm).

Step-by-step Solution:

1. Identify parameters: Image Size = 60 mm; Magnification = ×600.
2. State formula for Actual Size: **Actual Size = Image Size / Magnification**
3. Calculate Actual Size in mm: **Actual Size = 60 mm / 600 = 0.1 mm**
4. Convert mm to micrometres (µm): **0.1 mm × 1,000 = 100 µm**
5. Final Answer: **Actual length of cell = 100 µm**

9.7 Revision Questions — Chapter 9

1. State six rules for making standard biological drawings.
2. Differentiate between scientific drawing and artistic drawing.
3. Convert 0.08 millimetres into micrometres.
4. An organism measures 0.5 mm in actual length. A student draws it 50 mm long. Calculate the magnification.
5. A leaf cell drawing measures 45 mm under ×300 magnification. Calculate the actual cell size in µm.

10. INTRODUCTION TO BIOLOGY PRACTICAL SKILLS

10.1 Core Scientific Methods and Skills

Biology relies on scientific inquiry and laboratory investigation. Essential practical skills include:

- **Observation:** Using senses and scientific instruments (magnifiers, microscopes) to gather detailed qualitative and quantitative data about specimens.

- **Measurement:** Quantifying biological variables accurately using standardized equipment (rulers, measuring cylinders, stopwatches, balances).
- **Classification:** Sorting biological samples into logical groups based on observable shared characteristics.
- **Recording Results:** Presenting collected data in clear written notes, structured text summaries, or labelled diagrams.

10.2 Scientific Method Concepts

Hypothesis

A testable, logical proposal or prediction put forward to explain a scientific observation. For example: "Higher water temperatures increase the rate of diffusion of solute molecules."

Variables

Factors or conditions that can change during an experiment:

- **Independent Variable:** The factor deliberately changed or manipulated by the investigator (e.g., water temperature).
- **Dependent Variable:** The factor measured to determine the effect of the independent variable (e.g., time taken for solute to diffuse).
- **Controlled Variables:** Factors kept strictly constant to ensure a fair test (e.g., volume of liquid, solute concentration, vessel size).

Fair Test, Accuracy, Precision, and Reliability

- **Fair Test:** An experiment where only the independent variable is changed, while all controlled variables remain strictly constant.
- **Accuracy:** How close a experimental measurement is to its true actual value.
- **Precision:** How close repeated measurements of the same sample are to one another.
- **Reliability:** Consistency of experimental results when trials are repeated multiple times. High reliability is achieved through experimental repetition.

10.3 Laboratory Safety Rules

- Wear protective lab coats and safety goggles during experiments involving chemicals or heating.
- Never eat, drink, or taste any chemical or biological sample inside the laboratory.
- Handle scalpels, glass slides, and mounted needles carefully to prevent cuts.
- Point test tubes away from yourself and others when heating chemicals over a Bunsen flame.
- Dispose of biological samples, chemical wastes, and glass fragments in designated disposal containers.
- Wash hands thoroughly with soap and water before leaving the biology laboratory.

10.4 Revision Questions — Chapter 10

1. Define an independent variable and a dependent variable in a biological experiment.
2. Explain why repeating experimental trials improves data reliability.
3. Differentiate between accuracy and precision in scientific measurements.
4. What is a hypothesis? Give an example of a biological hypothesis.
5. List five safety precautions to observe in a senior secondary biology laboratory.

11. FIRST TERM PRACTICAL ACTIVITIES

11.1 Practical Activity 1: Observing Onion Epidermal Cells

Aim: To prepare a temporary wet mount of onion epidermal cells and observe plant cell structures under a light microscope.

Materials: Fresh onion bulb, scalpel, forceps, glass slides, cover slips, iodine solution, dropper, tissue paper, light microscope.

Procedure:

1. Cut open an onion bulb and peel a thin, transparent layer of inner epidermis using forceps.
2. Place a small drop of water on a clean glass slide and lay the thin onion strip flat without folding.
3. Add one drop of iodine stain solution to highlight cell nuclei and cell walls.
4. Gently lower a cover slip at a 45-degree angle using a mounted needle to avoid trapping air bubbles.
5. Blot excess stain using tissue paper and observe under low power, then switch to high power magnification.

Expected Observation: Tightly packed, brick-like rectangular cells. Distinct cellulose cell walls, cytoplasmic matrix, and stained cell nuclei are visible.

Precautions: Ensure onion layer is flat without folds. Avoid air bubbles under the cover slip.

Student Questions: (1) Why is iodine stain added to the slide? (2) State two cell structures visible in onion cells that are absent in animal cells.

11.2 Practical Activity 2: Investigating Osmosis Using Potato Tissue

Aim: To demonstrate osmosis in living plant tissue using potato cylinders placed in different salt solution concentrations.

Materials: Fresh Irish potato, cork borer, ruler, razor blade, electronic balance, beaker 1 (distilled water), beaker 2 (20% concentrated salt solution), paper towels, stopwatch.

Procedure:

1. Use a cork borer to prepare six uniform potato cylinders. Cut each cylinder to exactly 40 mm in length.
2. Measure and record initial mass and length of each potato cylinder.
3. Place three cylinders into Beaker 1 (distilled water) and three cylinders into Beaker 2 (20% salt solution).
4. Leave setups undisturbed for 60 minutes.
5. Remove potato cylinders, blot gently with paper towels, and re-measure length and mass.

Expected Observation: Cylinders in distilled water gain mass and length, becoming firm and turgid. Cylinders in concentrated salt solution lose mass and length, becoming soft and plasmolysed (flaccid).

Explanation: Water molecules diffuse into potato cells in distilled water by osmosis down a water potential gradient. In salt solution, water diffuses out of potato cells into the hypertonic solution.

Sources of Error: Incomplete blotting of surface liquid; variations in initial potato cylinder diameter.

Student Questions: (1) Why did potato cylinders in distilled water become turgid? (2) Calculate the percentage change in mass for cylinders in salt solution.

11.3 Practical Activity 3: Demonstrating Diffusion in Liquids

Aim: To observe and demonstrate the rate of diffusion of potassium permanganate crystals in water.

Materials: 250 ml glass beaker, distilled water, potassium permanganate crystals, forceps, stopwatch.

Procedure:

1. Fill a glass beaker with 200 ml of undisturbed distilled water and place on a flat table.
2. Using forceps, gently drop a single crystal of potassium permanganate into the center of the beaker bottom.
3. Do not stir or disturb the beaker. Observe purple color movement at 5-minute intervals.

Expected Observation: Intense purple color gradually spreads outward from the crystal base throughout the liquid until uniform purple color is achieved.

Explanation: Potassium permanganate particles dissolve and move passively from a region of high concentration at the crystal base to lower concentration throughout water.

11.4 Revision Questions — Chapter 11

1. Why is a cover slip lowered at an angle during wet mount slide preparation?
2. Explain why potato cylinders left in concentrated salt solution become soft and shrink.
3. What safety precaution must be taken when handling scalpels and glass slides in practical sessions?
4. Identify two sources of experimental error in the potato osmosis practical activity.

12. COMMON BIOLOGY MISCONCEPTIONS

12.1 Detailed Deconstruction of Misconceptions

MISCONCEPTION 1: RESPIRATION IS THE SAME AS BREATHING.

Why it is Incorrect: Breathing is simply the physical mechanism of gas exchange across respiratory surfaces. Respiration is an intracellular biochemical process occurring inside mitochondria to yield ATP energy from glucose oxidation.

MISCONCEPTION 2: PLANTS DO NOT RESPIRE; THEY ONLY PERFORM PHOTOSYNTHESIS.

Why it is Incorrect: Living plant cells require ATP energy continuously 24 hours a day to perform metabolic activities. Plants respire continuously day and night, absorbing oxygen and releasing carbon dioxide in mitochondria.

MISCONCEPTION 3: PLANTS PHOTOSYNTHESIZE DURING THE DAY AND RESPIRE ONLY AT NIGHT.

Why it is Incorrect: Photosynthesis occurs only during daylight hours because it requires light energy. However, cellular respiration occurs continuously day and night without stopping. During daylight, photosynthesis rate exceeds respiration rate.

MISCONCEPTION 4: PLANTS DO NOT MOVE.

Why it is Incorrect: While plants lack locomotion to move their whole body from place to place, plants exhibit tropic growth movements (phototropism, geotropism) and organ movements (leaf folding in *Mimosa* species).

MISCONCEPTION 5: ALL CELLS POSSESS A CELL WALL.

Why it is Incorrect: Animal cells and protozoans completely lack cell walls. Cell walls are present only in plant cells (cellulose), fungal cells (chitin), and bacterial cells (peptidoglycan).

MISCONCEPTION 6: ALL PLANT CELLS CONTAIN CHLOROPLASTS.

Why it is Incorrect: Chloroplasts occur only in green photosynthetic plant tissues (such as leaf palisade mesophyll). Underground root cells and inner stem cells lack chloroplasts and do not undergo photosynthesis.

MISCONCEPTION 7: OSMOSIS IS THE MOVEMENT OF SOLUTE OR SALT PARTICLES.

Why it is Incorrect: Osmosis refers strictly to the passive net movement of solvent (water) molecules across a selectively permeable membrane down a water potential gradient.

MISCONCEPTION 8: ALL MICROORGANISMS ARE HARMFUL PATHOGENS.

Why it is Incorrect: Most microorganisms are beneficial or essential for life. Decomposing bacteria and fungi recycle soil nutrients, gut microbiota aid digestion, and industrial yeast fermented products support economy.

MISCONCEPTION 9: GROWTH SIMPLY MEANS GETTING BIGGER IN SIZE.

Why it is Incorrect: True biological growth involves a permanent increase in cell number, dry mass, and cellular complexity through mitosis and differentiation, not temporary swelling from water intake.

12.2 Revision Questions — Chapter 12

1. Refute the claim that green plants only respire at night.
2. Explain why non-green plant tissues such as root cells do not contain chloroplasts.
3. Clarify the biological error in stating that osmosis is "the movement of salt into cells".
4. Give three examples of beneficial activities performed by microorganisms in everyday life.

13. FIRST TERM REVISION QUESTIONS & EXAMINATIONS

13.1 Part A: 30 Multiple-Choice Questions

1. The study of the structure and function of biological cells is known as:

A. Ecology B. Cytology C. Taxonomy D. Physiology

2. Which characteristic of life involves removing toxic metabolic waste products from cells?

A. Egestion B. Nutrition C. Excretion D. Respiration

3. Which organelle is referred to as the powerhouse of the cell?

A. Nucleus B. Chloroplast C. Ribosome D. Mitochondrion

4. The binomial scientific name of an organism consists of:

A. Family and Genus B. Genus and Species C. Order and Family D. Phylum and Species

5. Which of the following organisms belongs to Kingdom Monera?

A. *Amoeba proteus* B. *Rhizopus stolonifer* C. *Escherichia coli* D. *Euglena viridis*

6. A red blood cell placed in pure distilled water will experience:

A. Plasmolysis B. Crenation C. Haemolysis D. No change

7. The net movement of water molecules across a selectively permeable membrane down a water potential gradient is:

A. Diffusion B. Active transport C. Osmosis D. Transpiration

8. Which cellular structure controls all cell metabolic activities and carries DNA?

A. Cytoplasm B. Nucleus C. Golgi apparatus D. Lysosome

9. Cell walls of green plants are composed of:

A. Chitin B. Peptidoglycan C. Cellulose D. Glycogen

10. What is the total magnification of a light microscope using a ×10 eyepiece lens and a ×40 objective lens?

A. ×50 B. ×400 C. ×300 D. ×40

11. Plant cell organelles responsible for photosynthesis are:

A. Mitochondria B. Ribosomes C. Chloroplasts D. Vacuoles

12. The structural level of organization of a heart is a/an:

A. Tissue B. Cell C. Organ D. System

13. Active transport differs from diffusion because active transport requires:

A. Water B. Metabolic energy (ATP) C. High temperature D. Sunlight

14. Which scientist first observed living, moving single-celled organisms under a microscope?

A. Robert Hooke B. Antonie van Leeuwenhoek C. Theodor Schwann D. Rudolf Virchow

15. Fungal cell walls are composed of:

A. Cellulose B. Chitin C. Starch D. Protein

16. The fundamental basic unit of biological classification is the:

A. Genus B. Kingdom C. Species D. Phylum

17. Structural adaptation of palisade mesophyll cells for photosynthesis includes:

A. Biconcave shape B. High concentration of chloroplasts C. Long flagellum D. Rigid chitin wall

18. Converting 4.5 millimetres (mm) into micrometres (µm) yields:

A. 45 µm B. 450 µm C. 4,500 µm D. 45,000 µm

19. In an experiment testing how light intensity affects plant growth, light intensity is the:

A. Dependent variable B. Independent variable C. Controlled variable D. Continuous error

20. Which of the following is an example of an organ system in plants?

A. Palisade layer B. Root hair C. Shoot system D. Leaf blade

21. The shrinkage of plant cytoplasm away from the cell wall in hypertonic solution is:

A. Haemolysis B. Plasmolysis C. Turgidity D. Diffusion

22. Which kingdom includes prokaryotic organisms?

A. Monera B. Protista C. Fungi D. Plantae

23. Which organelle synthesizes lipids and detoxifies drugs?

A. Rough ER B. Smooth ER C. Lysosome D. Ribosome

24. What happens to a plant cell placed in a hypotonic solution?

A. Plasmolyses B. Becomes turgid C. Bursts (lyses) D. Shrivels

25. The correct biological term for eliminating undigested food faeces is:

A. Excretion B. Egestion C. Respiration D. Secretion

26. Which structure contains hydrolytic enzymes for intracellular digestion?

A. Lysosome B. Golgi body C. Nucleolus D. Ribosome

27. Which rule applies when handwriting a scientific name in an exam?

A. Capitalize species name B. Underline genus and species separately C. Enclose in quotes D. Write in bold

28. Membrane surrounding the large central vacuole in plant cells is called:

A. Plasma membrane B. Tonoplast C. Cell wall D. Nuclear envelope

29. The capacity of a microscope to distinguish two adjacent points as separate is:

A. Magnification B. Resolution C. Focal distance D. Contrast

30. Carbohydrates are stored in animal cells as:

A. Starch B. Cellulose C. Glycogen D. Sucrose

13.2 Answer Key & Explanations for Multiple-Choice Questions

- **1. B (Cytology):** Cytology is the branch of biology dedicated to studying cells.
- **2. C (Excretion):** Excretion is the removal of metabolic cellular waste.
- **3. D (Mitochondrion):** Mitochondria produce ATP energy during cellular respiration.
- **4. B (Genus and Species):** Binomial nomenclature pairs generic name and specific epithet.
- **5. C (*Escherichia coli*):** *E. coli* is a bacterium belonging to Kingdom Monera.
- **6. C (Haemolysis):** Red blood cells lack cell walls and burst in hypotonic water.
- **7. C (Osmosis):** Osmosis is water movement across semi-permeable membranes.

- **8. B (Nucleus):** The nucleus houses DNA and regulates cellular activities.
- **9. C (Cellulose):** Plant cell walls are made of cellulose structural polysaccharides.
- **10. B (×400):** Total magnification = Eyepiece lens (10) × Objective lens (40) = ×400.
- **11. C (Chloroplasts):** Chloroplasts contain chlorophyll for trapping solar energy.
- **12. C (Organ):** The heart contains cardiac muscle, nerve tissue, and connective tissue.
- **13. B (Metabolic energy - ATP):** Active transport moves solutes against concentration gradient using ATP.
- **14. B (Antonie van Leeuwenhoek):** Leeuwenhoek first observed living microscopic "animalcules".
- **15. B (Chitin):** Fungi cell walls consist of tough chitin polymer.
- **16. C (Species):** Species is the basic fundamental unit of biological classification.
- **17. B (High concentration of chloroplasts):** Palisade cells specialize in light capture.
- **18. C (4,500 µm):** $4.5 \text{ mm} \times 1,000 = 4,500 \text{ µm}$.
- **19. B (Independent variable):** The variable manipulated by the experimenter.
- **20. C (Shoot system):** Shoot system comprises stem, leaves, flowers, and fruits working together.
- **21. B (Plasmolysis):** Shrinkage of plant cell cytoplasm due to hypertonic water loss.
- **22. A (Monera):** Monera contains all prokaryotic organisms (bacteria).
- **23. B (Smooth ER):** Smooth ER synthesizes lipids and detoxifies chemicals.
- **24. B (Becomes turgid):** Water enters plant cell; cellulose wall prevents bursting.
- **25. B (Egestion):** Egestion is removing undigested food residual through anus.
- **26. A (Lysosome):** Lysosomes store hydrolytic enzymes for intracellular breakdown.
- **27. B (Underline genus and species separately):** Rule for handwritten scientific names.
- **28. B (Tonoplast):** Tonoplast is the selective membrane enclosing plant central vacuole.
- **29. B (Resolution):** Resolving power separates two close objects clearly.
- **30. C (Glycogen):** Animals store excess carbohydrates as liver and muscle glycogen.

13.3 Part B: Short-Answer Questions & Model Answers

Question 1: Define Taxonomy and state two advantages of classifying living organisms.

Model Answer: Taxonomy is the branch of biology concerned with the identification, naming, and classification of living organisms into hierarchical groups. Two advantages are: (1) It organizes vast biological diversity into structured categories for easier study, and (2) It establishes universal scientific names that facilitate clear global scientific communication.

Question 2: State three functions of the plant cell wall.

Model Answer: (1) Provides structural support and mechanical strength to keep the plant upright against gravity. (2) Protects internal protoplasm from physical damage and pathogen entry. (3) Prevents plant cells from bursting when taking in water by osmosis.

Question 3: Distinguish between diffusion and active transport.

Model Answer: Diffusion is a passive movement of particles from high to low concentration down a concentration gradient without requiring ATP energy. Active transport is an active process that moves solute particles from low to high concentration against a concentration gradient, requiring cellular ATP energy and carrier proteins.

Question 4: What is plasmolysis? How can a plasmolysed cell be reversed?

Model Answer: Plasmolysis is the shrinking of cytoplasm away from the cell wall of a plant cell when placed in a hypertonic solution due to water loss by osmosis. It can be reversed by placing the plasmolysed cell into a hypotonic solution (or pure water), causing water to re-enter by osmosis (deplasmolysis).

Question 5: Explain why viruses are considered obligate intracellular parasites.

Model Answer: Viruses are non-living outside host cells because they lack cellular structures, organelles, and metabolic machinery. They can only replicate and carry out life processes by invading and hijacking the metabolic machinery of a living host cell.

13.4 Part C: Structured Questions & Model Answers

Structured Question 1: Relative to Cell Structure & Transport

A student placed identical cylinders of raw yam measuring 50 mm in length into three test tubes containing different solutions (A, B, and C). After 2 hours, the length of the cylinder in Solution A was 54 mm, in Solution B was 50 mm, and in Solution C was 44 mm.

(a) Identify the nature of Solutions A, B, and C relative to yam cell sap.

Model Answer: Solution A is Hypotonic relative to cell sap (caused water entry and elongation). Solution B is Isotonic relative to cell sap (no net water movement, length unchanged). Solution C is Hypertonic relative to cell sap (caused water loss and shrinkage).

(b) Name the biological process responsible for the changes in length.

Model Answer: Osmosis.

(c) Explain the cellular change that occurred in the yam cylinder placed in Solution C.

Model Answer: Yam cells lost water by osmosis because Solution C had a lower water potential than the cell sap. The cells shrank and cytoplasm pulled away from cell walls (plasmolysis), causing the overall tissue length to decrease.

13.5 Part D: Application & Scenario-Based Questions

Scenario Question 1: Agricultural Application of Osmosis

A farmer in Jos applied an excessively heavy dose of dry chemical fertilizer around the roots of his growing tomato crops during a dry period. Two days later, he noticed that the tomato plants wilting rapidly despite receiving sunshine. Explain biologically why the plants wilted after fertilizer application.

Model Answer: Applying excess chemical fertilizer dissolved in soil moisture, creating an extremely concentrated (hypertonic) soil solution around the root environment with a very low water potential. Because the soil water potential became lower than the water potential inside the tomato root hair cells, water moved out of root hair cells into the soil by osmosis. This loss of water reduced turgor pressure in leaf and stem cells, causing cells to become plasmolysed and flaccid, resulting in rapid wilting of the plants.

14. GLOSSARY OF KEY BIOLOGICAL TERMS

Active Transport: Energy-requiring movement of substances across a cell membrane against a concentration gradient using ATP and protein carriers.

Anatomy: The branch of biology studying the internal physical structure of living organisms.

Autotrophic Nutrition: Mode of nutrition in which an organism manufactures organic food from inorganic raw materials using light or chemical energy.

Binomial Nomenclature: System of naming species using two Latinized terms representing Genus and Species.

Botany: The scientific study of plants.

Cell: The fundamental structural and functional unit of life.

Cell Membrane: Selectively permeable phospholipid bilayer enclosing cytoplasm of living cells.

Cell Wall: Rigid outer protective boundary surrounding plant, fungal, or bacterial cells.

Chloroplast: Double-membrane organelle in plant cells containing chlorophyll, site of photosynthesis.

Cytology: The scientific study of cells.

Cytoplasm: Aqueous jelly-like cellular matrix outside nucleus containing cell organelles and dissolved metabolites.

Diffusion: Passive net movement of particles from high to low concentration down a concentration gradient.

Ecology: Branch of biology studying interactions between living organisms and physical environment.

Eukaryotic Cell: Complex cell containing a true membrane-bound nucleus and specialized membrane-enclosed organelles.

Excretion: Removal of metabolic waste products, toxic substances, and excesses from living organisms.

Genetics: The scientific study of heredity and variation in living organisms.

Hypotonic Solution: Solution with lower solute concentration (higher water potential) than cell cytoplasm.

Hypertonic Solution: Solution with higher solute concentration (lower water potential) than cell cytoplasm.

Isotonic Solution: Solution having equal solute concentration (equal water potential) to cell cytoplasm.

Magnification: Factor by which an image is enlarged compared to actual physical size of specimen.

Mitochondrion: Double-membrane cell organelle responsible for aerobic respiration and ATP energy production.

Nucleus: Membrane-enclosed organelle containing genetic DNA controlling cellular activities.

Nutrition: Process of absorbing or synthesizing food nutrients required for life and growth.

Organ: Group of specialized tissues working together to perform specific body functions.

Organelle: Specialized membrane-bound structure inside eukaryotic cells carrying out specific tasks.

Osmosis: Net movement of water molecules from high to low water potential across semi-permeable membrane.

Plasmolysis: Shrinkage of plant cell cytoplasm away from cell wall due to osmotic water loss in hypertonic environment.

Prokaryotic Cell: Primitive cell lacking true membrane-bound nucleus and membrane-enclosed organelles.

Respiration: Intracellular metabolic breakdown of organic nutrients (glucose) to release ATP energy.

Resolution: Minimum distance at which two distinct points can be distinguished clearly as separate.

Sensitivity (Irritability): Ability of living organisms to detect environmental stimuli and respond.

Species: Group of similar organisms capable of interbreeding to produce fertile offspring.

Taxonomy: Science of biological identification, nomenclature, and hierarchical classification.

Tissue: Group of similar specialized cells performing a common specific function.

Turgidity: State of plant cell firmness resulting from internal fluid turgor pressure against cell wall.

Vacuole: Membrane-bound cell storage sac containing liquid sap or food contents.

Zoology: Scientific study of animals.